S&P GLOBAL & CRISIL CAMPUS HACKATHON 2026

AI/NLP Financial Risk Engine &

Tactical Index Rebalancing

Module A: Tactical High-Frequency Stock Index Rebalancer

CANDIDATE INFORMATION

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College / Campus:	**NIT JAMSHEDPUR**
Track / Module:	**MODULE A (Tactical Index Rebalancing)**

SYSTEM HIGHLIGHTS AT A GLANCE

- **FinBERT NLP: P(pos) - P(neg) Sentiment Scoring [-1.0 to +1.0]**
- **Hybrid 6-Category Financial Event Taxonomy**
- **15-Ticker S&P 100 Live Entity Extraction**
- **Autonomous Rebalancing with Water-Filling Risk Bounds**

Markets move on text. We score it instantly



The Problem

Prices react in seconds to unstructured text: news headlines and social media. Static index weights can't.



The Solution

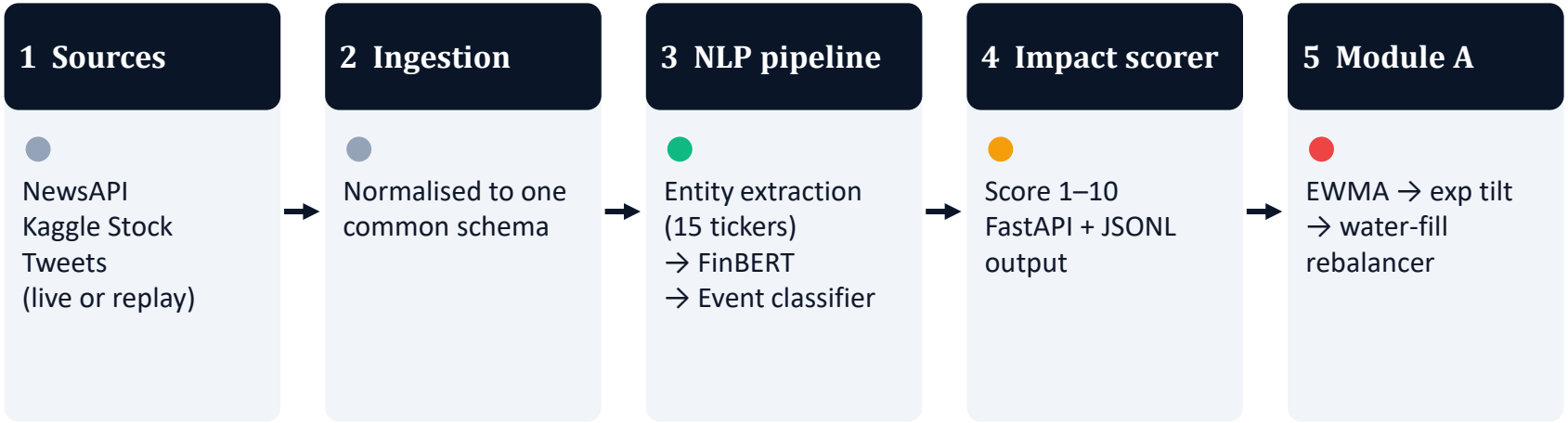
SentinelRisk quantifies textual impact in real time and tactically tilts index weights.



The Edge

Cuts exposure to negative-sentiment drawdowns while still capturing upside.

End-to-end architecture



NLP & impact scoring

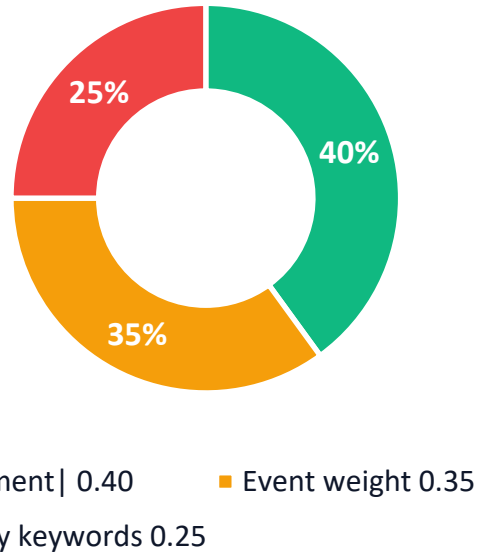
1. **Entity extractor**
   Alias dictionary + word-boundary regex

2. **Sentiment**
   ProsusAI/FinBERT: finance-tuned, batched

3. **Event classifier**
   Keyword rules first, DeBERTa zero-shot fallback

Impact = clip(round(10 × score), 1–10)

What drives the score



Module A: tactical index rebalancer

EWMA sentiment
per stock, configurable half-life

→

Exponential tilt
$w \propto \text{base} \times \exp(k \cdot \text{EWMA})$

→

Water-filling
enforces weight bounds

→

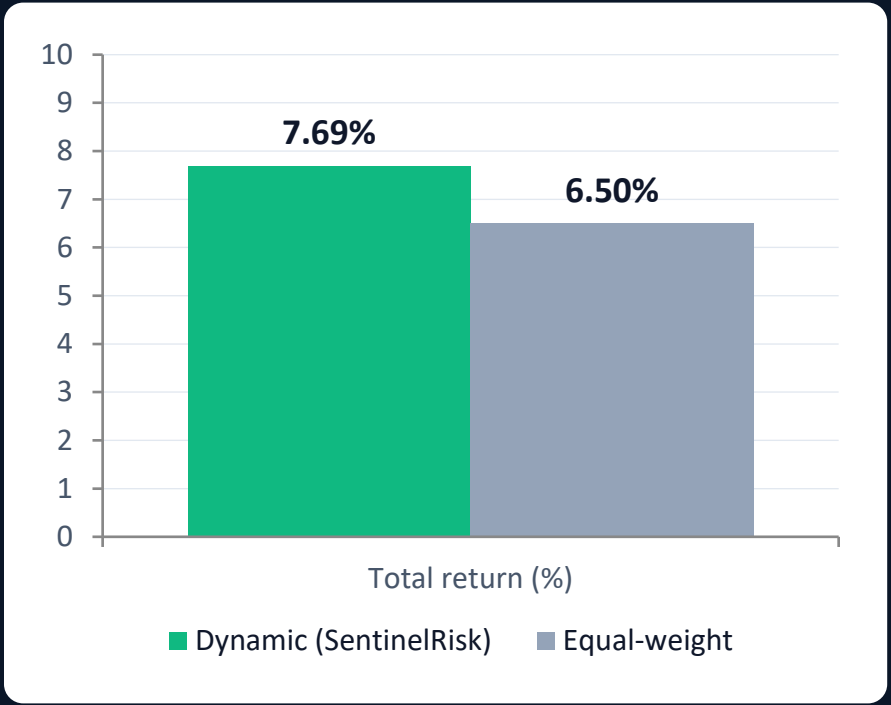
Apply T-1 to T
no look-ahead bias

Where each stock's weight is allowed to live (15-stock index)



5 bps transaction cost per unit of turnover is deducted in the backtest

Backtest: dynamic beats equal-weight



2.16
Sharpe ratio

14.69%
Annual volatility

-8.69%
Max drawdown

1.37%
Avg turnover

+1.19 pp over equal-weight, net of 5 bps costs

Built to survive messy real-world data

Latency	→	Batched sentiment inference
Ambiguity	→	Multi-ticker matching + MARKET fallback for general news
Noise	→	Hybrid event classifier ignores irrelevant text
Reliability	→	Offline mode, keyword sentiment, cached prices
Look-ahead bias	→	Strict chronological separation in the backtest

Roadmap: from prototype to platform



Module B

Stress tests triggered by events with Impact > 7

GDELT

Macro and geopolitical news coverage

Kafka

Streaming pipeline for real-time scale

Fine-tuned FinBERT

Trained on proprietary data

Convex optimisation

Advanced portfolio constraints

Turn headlines into risk control

FinBERT-powered signals • bounded, cost-aware rebalancing • no look-ahead

Sharpe 2.16 | +7.69%

Thank you. Questions?